Project Information

Richardson Residence
8560 Sand Point Way NE
Seattle, WA 98115

Contact Information

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Messages / Comments * **RESULT= PASS**

UA Reduction = 32.07, Proposed UA is better than baseline by 3%

Window area is 23% of floor area

-

Whole House Mechanical Ventilation Airflow Rate: 95 CFM with Run Time Percent of 100%, Balanced, Distributed

Maximum allowable total measured duct leakage: 519 CFM25

* Results assume your inputs are complete and correct. Results do not constitute an approval. Analysis should be reviewed by your AHJ.

ANALYSIS SET UP

What code compliance pathway are you using?	**U-Factor Compliance Path**
Project Building Type?	**New Construction**
Occupancy Type?	**R3 Single family dwellings and townhouses**
Code Version?	**WSEC 2021**
Classification:	Large Dwelling Unit -- 6486 sq. ft.
Baseline Description:	Code Baseline - Baseline and proposed window areas are equal.
About Your Selection:	Up to 15 sf exempt window and 24 sf exempt door allowable
	-

RESULTS - Comparison of Baseline and Proposed Design **

Component Performance, R occupancies	Baseline Design U *	Baseline Design Area	Baseline Design UA	Proposed Design U	Proposed Design Area	Proposed Design UA
Doors U =	0.300	121	36.3	0.300	121	36.3
Overhead Glazing U =	0.500	0	0.0		0	0.0
Vertical Glazing U =	0.300	1,499	449.6	0.300	1,499	449.6
Flat/Vaulted Ceilings U =	0.024	2,470	59.3	0.024	2,470	59.3
Wall (above grade) U =	0.056	3,166	177.3	0.054	3,166	171.0
Floors over Crawlspace U =	0.029	0	0.0		0	0.0
Slab on Grade F =	0.540	143	77.2	0.360	143	51.5
Below Grade Wall U =	0.035	1,164	40.7	0.035	1,164	40.7
Below Grade Slab F =	0.500	188	93.8	0.500	188	93.8

* Values from Table R402.1.2 (Oct 2023)

Baseline UA Total	934.1	**Proposed UA Total**	902.1	
Required Credits	9.0	**Proposed Credits**	9.0	from Tables 406.2 and 406.3
		UA Percent Reduction	3%	
		UA Reduction	32.1	

If the Proposed UA ≤ the Target UA, and the Proposed Credits from Table 406 are ≥ those required in Section R406, then the home meets the WSEC.

** Results assume your inputs are complete and correct. Results do not constitute an approval. Analysis should be reviewed by your AHJ.

Table R406.2 Energy Equalization Credits

System No.	Full Description	Select System Type	Fuel Normalization Credits (406.2)	Energy Credits (406.3)	Total Credits (406.2 & 406.3)
4	For heating system using a heat pump that meets federal standards for the equipment listed in Table C403.3.2(2) or Table C403.3.2(9) OR Air to Water heat pump units that are configured to provide both heating and cooling and are rated in accordance with AHRI 550/590	**Variable Refrigerant Heat Pump or Air-to-Water Heat Pump**	3.0	6.0	9.0

Table R406.3 Energy Credits

Option No.	Category	Select Options	Energy Credits	Brief Description of Selected Options*
1	Efficient Building Envelope	**Not Selected**	0.0	-
2	Air Leakage Control and Efficient Ventilation	**Option 2.2**	1.5	Per Section R402.4.1.2 /1.5 ACH50 / For R-2, 0.20 cfm per ft2 at 50 Pa. / HRV with min SHR eff of 0.75 per IRC Section M1505.3 or IMC Section 403.8
3.1 -3.10	High Efficiency HVAC	**Option 3.6**	1.0	Air source ducted Heat Pump w/ Min HSPF2 of 9.4 (HSPF of 11). A ducted cold climate variable capacity heat pump on the NEEP qualified product list with a min HSPF2 of 8.5 (HSPF of 10) may be used to satisfy this requirement. If htg design temp is 23F or below,a cold climate variable capacity heat pump is required.
3.11	High Efficiency HVAC: Smart Thermostat	**Not Selected**	NA	
4	High Efficiency HVAC Distribution System	**Option 4.1**	0.5	Ducts/distribution system in conditioned space per R403.3.2 Electric resistance, hydronic, ductless and gas fired systems < 80% AFUE systems not permitted

5.1	Efficient Water Heating: Drain Heat Recovery				0.0	
5.2	Efficient Water Heating: Compact Hot Water Distribution				0.0	
5.3-5.8	Efficient Water Heating			**Option 5.4**	1.0	Energy star gas or propane water heater with min UEF of 0.91 OR Solar supplemental water heating OR GSHP meeting requirements of Option 3.4
6	Renewable Electric Energy	2,400	kWh	**Option 6.1**	2.0	On-site wind or solar electric energy, 0.5 credit per 600 kWh of generation up to maximum of 4.5 credits
7	Appliance Package				0.0	
				Energy Credits	**6.0**	

***Refer to WSEC 2021 Table R406.3 for complete option descriptions and requirements** https://sbcc.wa.gov/state-codes-regulations-guidelines/state-building-code/energy-code

THERMAL ENVELOPE DETAILS - Proposed Design

Conditioned Floor Area, Proposed Design	6,486	sq. ft
Classification	Large Dwelling Unit	
Notes		

Exterior Doors

Plan ID	Component Description	Ref.	Door U	Qt.	Width Feet	Width Inch	Height Feet	Height Inch	Area	UA
Exempt									0	0.0
Lower	Code Baseline, U=0.30	-	0.30	1	3	6	8	0	28	8.4
Main	Code Baseline, U=0.30	-	0.30	1	5	0	9	0	45	13.5
Upper	Code Baseline, U=0.30	-	0.30	2	3	0	8	0	48	14.4
									0	0.0
									0	0.0
									0	0.0
									0	0.0
									0	0.0
									0	0.0
									0	0.0
								Sum of Area and UA	**121**	**36.3**
								Exterior Doors Area Weighted U		**0.300**

Overhead Glazing

Plan ID	Component Description	Ref.	Glazing U	Qt.	Width Feet	Width Inch	Height Feet	Height Inch	Area	UA
									-	
									-	
									-	
									-	
									-	
								Sum of Area and UA	**0.0**	**0**
								Overhead Glazing Area Weighted U		

Vertical Glazing Schedule

Rows to Show 32

	Plan ID	Component Description	Ref.	Glazing U	Qt.	Width Feet	Width Inch	Height Feet	Height Inch	Area	UA
	Exempt			-						-	-
1	Lower									-	-
2	Ex. Room	U=0.30 (Code Baseline)	Table 406.2	0.30	1	10	0	8	0	80.0	24.00
3	Rec Room	U=0.30 (Code Baseline)	Table 406.2	0.30	1	20	0	8	0	160.0	48.00
4	Rec Room	U=0.30 (Code Baseline)	Table 406.2	0.30	1	7	0	5	0	35.0	10.50
5										-	-
6										-	-
7	Prim. WC	U=0.30 (Code Baseline)	Table 406.2	0.30	1	2	0	5	6	11.0	3.30
8	Prim. #1	U=0.30 (Code Baseline)	Table 406.2	0.30	1	9	0	2	0	18.0	5.40
9	Prim. #1	U=0.30 (Code Baseline)	Table 406.2	0.30	1	9	0	9	9	87.8	26.33
10	Grt. Room	U=0.30 (Code Baseline)	Table 406.2	0.30	1	16	0	9	9	156.0	46.80
11	Din/Kitch	U=0.30 (Code Baseline)	Table 406.2	0.30	1	17	6	9	9	170.6	51.19
12	TV Rm.	U=0.30 (Code Baseline)	Table 406.2	0.30	1	9	0	6	0	54.0	16.20
13	Mud Rm.	U=0.30 (Code Baseline)	Table 406.2	0.30	1	3	0	8	0	24.0	7.20
14	Mud Rm.	U=0.30 (Code Baseline)	Table 406.2	0.30	1	3	0	6	6	19.5	5.85
15	Stairs	U=0.30 (Code Baseline)	Table 406.2	0.30	1	8	0	9	0	72.0	21.60
16	Pwdr.	U=0.30 (Code Baseline)	Table 406.2	0.30	1	2	0	5	6	11.0	3.30
17										-	-
18	Upper									-	-
19	Prim. #2 b	U=0.30 (Code Baseline)	Table 406.2	0.30	1	2	0	6	0	12.0	3.60
20	Prim. #2 W	U=0.30 (Code Baseline)	Table 406.2	0.30	1	2	0	4	0	8.0	2.40
21	Prim. #2	U=0.30 (Code Baseline)	Table 406.2	0.30	1	9	0	2	0	18.0	5.40
22	Prim. #2	U=0.30 (Code Baseline)	Table 406.2	0.30	1	14	0	8	9	122.5	36.75
23	Office	U=0.30 (Code Baseline)	Table 406.2	0.30	1	8	0	8	9	70.0	21.00
24	Bdrm #3/4	U=0.30 (Code Baseline)	Table 406.2	0.30	2	8	8	8	9	151.7	45.50
25	Bdrm #4/5	U=0.30 (Code Baseline)	Table 406.2	0.30	2	2	0	4	6	18.0	5.40
26	Lndry.	U=0.30 (Code Baseline)	Table 406.2	0.30	1	3	0	4	6	13.5	4.05
27										-	-
28	Bdrm #5	U=0.30 (Code Baseline)	Table 406.2	0.30	1	10	0	6	6	65.0	19.50
29	Hall	U=0.30 (Code Baseline)	Table 406.2	0.30	1	6	0	5	0	30.0	9.00
30	Hall	U=0.30 (Code Baseline)	Table 406.2	0.30	1	3	0	5	0	15.0	4.50
31	Stairs	U=0.30 (Code Baseline)	Table 406.2	0.30	1	8	0	8	0	64.0	19.20
32	Prim. #2 B	U=0.30 (Code Baseline)	Table 406.2	0.30	1	2	0	6	0	12.0	3.60
								Sum of Area and UA	**1,498.5**	**449.6**
								Vertical Glazing Area Weighted U		**0.300**
								Vertical Glazing and Doors Area Weighted U		**0.300**

Flat/Vaulted Ceilings

Plan ID	Component Description	Ref.	Attic U	Area	UA
Flat	2021 U-factor Baseline (Table R402.1.2)	Table R402.1	0.024	2,470	59.3
			Sum of Area and UA	2,470	59.3
			Area Weighted U-Value		0.024

Walls (Above Grade)

Plan ID	Component Description	Ref.	Wall U	Net Area	UA
Lower	R21 cavity+R0 foam INT 2X6W Lap (Code Baseline)	10-5	0.054	429	23
Main	R21 cavity+R0 foam INT 2X6W Lap (Code Baseline)	10-5	0.054	1,171	63
Upper	R21 cavity+R0 foam INT 2X6W Lap (Code Baseline)	10-5	0.054	1,566	85
			Sum of Net Area and UA	3,166	171
			Gross Wall Area	4,786	ft^2
			Area Weighted U-Value		0.054

Floor (over crawl or exterior)

Plan ID	Component Description	Ref.	Floor U	Area	UA
			Sum of Area and UA	0	0
			Area Weighted U-Value		

Slab on Grade (less than 2 feet below grade)

Plan ID	Component Description	Ref.	Slab F	Slab Perim	FP
	R10 Fully insulated (2018 1.3-1.5)	10-2	0.360	143	51
			Sum of Perimeter and FP	143	51
			Area Weighted U-Value		0.360

Below Grade Walls and Slabs

Plan ID	Component Description	Slab Depth	Ref.	Wall U	Wall Area	Wall UA	Slab F	Slab Perim	Slab UA
	R21 Batt w/TB	7' depth	WSU	0.035	1,164	40.7	0.500	188	94
			Sum of Area, Length and UA		1,164	41		188	94
			Weighted U- and F-values			0.035			0.500

Links to Download Forms, Checklists and Other Resources	Link	
Compliance Certificate	Compliance Certificate	Instructions
Insulation Certificate for Residential New Construction	Insulation Certificate	
Duct Testing Affadavits		
Existing Construction	Affidavit, Existing	
New Construction	Affidavit, New	
Prescriptive Checklist for 2018 WSEC	Prescriptive Checklist	
Alterations (Remodel) Worksheet	Worksheet	
EER SEER2 COP HSPF2 Converter	https://www.adicotengineering.com/eer-seer2-cop-hspf2-kwton-converter	

Show Ventilation Calculator?	Show	

Ventilation Requirements

Conditioned Floor Area	6,486 sq. ft.	
Number of Bedrooms	5	
Run-Time Percent in Each 4-Hour Segment	100%	
Is the system Balanced?	Balanced	Verify system meets definition of 'Balanced Whole-House Ventilation'
Is the system Distributed?	Distributed	Verify system meets definition of 'Distributed Whole-House Ventilation'
Ventilation Code Section	IRC, Chapter 15	
Whole House Mechanical Ventilation Airflow Rate	95 CFM	

Show Distribution System Calculator?	Show	

HVAC Thermal Distribution System — Download RS-33 (2018) http://www.energy.wsu.edu/Documents/Duct%20Testing%20Standards%20_2018%20V

Is this a hydronic heating system?	No
Location of Ducts	Conditioned Space
Location of Air Handler	Conditioned Space
Is Duct Testing Required?	Yes

Maximum Duct Leakage:

Maximum total measured duct leakage per square foot	0.08	CFM25 per sq. ft.
Maximum allowable total measured duct leakage	519	CFM25

A maximum of 10 feet of return ducts and 5 feet of supply ducts are allowed to be located outside of the building thermal envelope, if insulated and sealed per R403.3.7.

Show Heating System Sizing?	Show	

Heating System Sizing - Proposed Design — Try Out BetterBuiltNW's HVAC Sizing Tool: https://betterbuiltnw.com/resources/hvac-sizing-tool

Nearest Weather Station	Seattle: Sea-Tac AP		
Indoor Design Temperature	70	F	
Outdoor Design Temperature	24	F	
Design Temperature Difference (ΔT)	46	F	
Conditioned Floor Area, Proposed Design	6,486	ft2	
Conditioned Volume Leave blank to use default of 8.5 ft. ceiling height	55,131	ft3	Average ceiling height =8.5 ft. Volume = 55131 ft3
Average ceiling height	8.5	ft	
HVAC System Type	Heat Pump		
Location of HVAC Distribution System	Conditioned Space		
Sum of UA, including exempt door and window	902		
Envelope Heat Load Sum of UA X ΔT	41,496	Btu / Hour	
Air Leakage Heat Load ((Volume X 0.6) X ΔT) X .018))	27,389	Btu / Hour	
Building Design Heat Load Air Leakage + Envelope Heat Loss	68,885	Btu / Hour	
Building and Duct Heat Load For ducts located in unconditioned space: Sum of Building Heat Loss X 1.1 For ducts located in conditioned space or ductless: Sum of Building Heat Loss X 1	68,885	Btu / Hour	
Maximum Heat Equipment Output Building and Duct Heat Loss X 1.25 for heat pumps Building and Duct Heat Loss X 1.40 for all other systems	86,106 25.2	Btu / Hour kW	